

Young-Seok Lee

Lab | [Google Scholar](#)

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ABOUT ME

- **Ph.D. Candidate** in Engineering from *The Electrical and Computer Engineering Department at Seoul National University*
- Supervisor: **Prof. Emeritus Sangwook Nam**(IEEE Life Senior Member) and **Prof. Jungsuek Oh**(IEEE Senior Member)
- Research interests: *RF System Design, RF Wireless Power Transfer, Optimization Algorithm, Large Scale Antenna Calibration and Operation, Target Detection, and Indoor Localization.*

EDUCATION

Seoul National University

Ph.D., Dept. of Electrical and Computer Engineering

Seoul, South Korea

Sep. 2023 – Present

- PI (Co-advised): Prof. Sangwook Nam (Ph.D. under Prof. Tatsuo Itoh)
- PI (Co-advised): Prof. Jungsuek Oh (Ph.D. under Prof. Kamal Sarabandi)
- Dissertation: TBD

Seoul National University

M.S., Dept. of Electrical and Computer Engineering

Seoul, South Korea

Sep. 2021 – Aug. 2023

- PI: Prof. Sangwook Nam
- Thesis: Efficient Wireless Power Transfer System Using the Received Power Feedback Information

Korea University

B.S., The School of Electrical Engineering

Seoul, South Korea

Mar. 2017 – Aug. 2021

- Advisor: Prof. Chulwoo Kim (Ph.D. under Prof. Sung Mo Kang)
- Thesis: High Speed Two-stage Folded Cascode Op-amp

Daejeon Dongsin Science High School (DDSHS)

Science High School in S. Korea

Daejeon, South Korea

Mar. 2014 – Feb. 2017

PUBLICATIONS

> Symbol '*' denotes 'corresponding author(s)'

> Symbol '=' denotes 'equal contributions'

International Journals(Submitted/Under review)

- [2] **Y.-S. Lee(=)**, S. Kim(=), T. Yoon, S. Nam*, and J. Oh* "Virtual-Beacon Depth Refocusing for Improved Received-Power Uniformity in Near-Field Time-Reversal Wireless Power Transfer," *IEEE Antennas and Wireless Letters (AWPL)*, Submitted.
- [1] T. Yoon, **Y.-S. Lee**, M. Kim, S. Lee, J. Lee, S. Nam, and J. Oh*, "Amplitude Tapering-Aware Efficient Near-Field Array Wireless Power Transfer Using Multi-State Supply Switching," *IEEE Open Journal of Antennas and Propagation (OJAP)*, Major Revision.

International Journals(First author)

- [3] **Y.-S. Lee**, T. Yoon, J. Oh, and S. Nam*, "An End-to-End DC-to-DC Efficiency Optimization and Design Analysis for MIMO Wireless Power Transfer Systems," *IEEE Transactions on Microwave Theory and Techniques (TMTT)*, Early Access.
- [2] **Y.-S. Lee**, T. Yoon, Y. J. Song, S. K. Hong, J. Oh, and S. Nam*, "Transmission-Conversion Efficiency Maximization Technique for MIMO Wireless Power Transfer Systems in ISAC Applications," *IEEE Antennas and Wireless Letters (AWPL)*, vol. 24, no. 11, pp. 4492-4496, Nov. 2025.
- [1] **Y.-S. Lee**, T. Yoon, M. Kim, S. Lee, B. Jung, J. Oh, and S. Nam*, "A Design and Characterization Method of a Scalable Large Transmitting Array for Wireless Power Transfer," *IEEE Transactions on Microwave Theory and Techniques (TMTT)*, vol. 73, no. 6, pp. 3346-3358, Jun. 2025.

International Journals(Co-author)

- [4] T. Yoon, **Y.-S. Lee**, U. Park, M. Kim, S. Lee, S. Nam, and J. Oh*, "Hardware-Aware MILP for Optimal End-to-End Efficiency in MISO WPT: Joint Optimization of Power Modes and Fault Tolerance," *IEEE Transactions on Microwave Theory and Techniques (TMTT)*, Accepted.

- [3] S. Kim, **Y.-S. Lee**, and S. Nam*, “Wideband Dual-Polarized End-Fire Slit Horn Antenna with Tightly Coupled Dipole and Monopole Probe Arrays,” *IEEE Transactions on Antennas and Propagation (TAP)*, vol. 73, no. 11, pp. 8528-8535, Nov. 2025.
- [2] T. Yoon, U. Park, M. Kim, S. Lee, **Y.-S. Lee**, S. Nam, and J. Oh*, “A Topology-based Array Compensation Empowered by Equivalent Current Modeling of TM10 Patch Antennas for Cross-Polarization Reduction,” *IEEE Open Journal of Antennas and Propagation (OJAP)*, vol. 6, no. 6, pp. 1696-1707, Dec. 2025.
- [1] H.-Y. Kim, **Y.-S. Lee**, and S. Nam*, “Efficiency Bound Estimation for a Practical Microwave and mmWave Wireless Power Transfer System Design,” *Journal of Electromagnetic Engineering and Science (JEES)*, vol. 23, no. 1, pp. 69-74, Jan. 2023.

International Conferences

- [8] **Y.-S. Lee**, T. Yoon, S. Nam, and J. Oh*, “DC-to-DC Efficiency Optimization in Wireless Power Transfer with PA and Rectifier Nonlinearities via Semidefinite Relaxation,” *2026 IEEE Wireless Power Technology Conference and EXPO (WPTCE 2026)*, Halifax, NS, Canada, Jul. 4-7, 2026. **(Best Student Paper Award, Sole Recipient(1st Place))**
(Selected for IEEE MTT-S Student Travel Grant)
- [7] **Y.-S. Lee**, T. Yoon, J. Oh, and S. Nam*, “Design and Calibration of a CW Time Reversal Wireless Power Transfer Transmitter with Pilot Signal Detection,” *2025 Asia-Pacific Microwave Conference (APMC 2025)*, Jeju, South Korea, Dec. 2-5, 2025. **(Invited Paper for Prof. Sangwook Nam and presented by Young-Seok Lee)**
- [6] **Y.-S. Lee**, J. Oh, and S. Nam*, “Effects of Magnitude Dynamic Range Constraints on MIMO Wireless Power Transfer Efficiency,” *2025 International Symposium on Antennas and Propagation (ISAP 2025)*, Fukuoka, Japan, Oct. 27-31, 2025.
- [5] T. Yoon, **Y.-S. Lee**, M. Kim, S. Lee, J. Lee, S. Nam, and J. Oh*, “Overcoming Efficiency Degradation in Wireless Power Transfer Systems: A Supply Voltage Modulation Method Empowered by 5.64-GHz 256-Element Antenna Array Receiving 10.6-Watt,” *2025 IEEE MTT-S International Microwave Symposium (IMS 2025)*, San Francisco, CA, USA, Jun. 15-20, 2025.
- [4] **Y.-S. Lee**, J. Oh, and S. Nam*, “Sequential Feedback-Based Phase Optimization Using Hadamard Basis for Wireless Power Transfer,” *2025 IEEE Wireless Power Technology Conference and EXPO (WPTCE 2025)*, Rome, Italy, Jun. 3-6, 2025. **(Selected for Ph.D. Students Initiative Program)**
- [3] **Y.-S. Lee**, J. Oh, and S. Nam*, “An Effect of Time Reversal Based Multiple Beacon Selection on Wireless Power Transfer Performance,” *2024 International Symposium on Antennas and Propagation (ISAP 2024)*, Incheon, South Korea, Nov. 5-8, 2024. **(Best Paper Award, 2nd Prize)**
- [2] **Y.-S. Lee**, T. Yoon, S. Lee, M. Kim, J. Lee, J. Oh, and S. Nam*, “LUT-Based Transmit Mode Calibration Complexity Reduction Method for Wireless Power Transfer,” *2024 IEEE Wireless Power Technology Conference and EXPO (WPTCE 2024)*, Kyoto, Japan, May 8-11, 2024. **(Selected for Student Travel Grant)**
- [1] **Y.-S. Lee** and S. Nam*, “Test Count Reduction Algorithm for Determining the Optimal Phase in Wireless Power Transfer using Bayesian Optimization,” *XXXVth URSI General Assembly and Scientific Symposium (URSI/GASS 2023)*, Sapporo, Japan, Aug. 19-26, 2023.

Domestic Conferences

* KIEES: The Korean Institute of Electromagnetic Engineering and Science, the primary society for electromagnetics in Korea.

- [6] **Y.-S. Lee**, T. Yoon, S. Nam, and J. Oh*, “End-to-End DC Power Optimization in Wireless Power Transfer Systems,” *2026 KIEES Winter Conf.*, Feb. 2026.
- [5] T. Yoon, **Y.-S. Lee**, and J. Oh*, “MILP-based Optimal Beamforming in MISO WPT System: Dealing with Nonlinear Power Consumption and Discrete Voltage Rails,” *2026 KIEES Winter Conf.*, Feb. 2026. **(Best Paper Award, IEEE MTT-S Seoul Chapter Award)**
- [4] **Y.-S. Lee**, T. Yoon, J. Oh, and S. Nam*, “Received DC Power Maximization in a Practical Wireless Power Transfer Systems,” *2025 KIEES Winter Conf.*, Feb. 2025. **(Best Paper Award, Grand Prize(1st Place))**
- [3] T. Yoon, **Y.-S. Lee**, M. Kim, S. Lee, and J. Oh*, “Comprehensive Optimization of Beamforming Wireless Power Transfer: Achieving 500 W Power Saving through Hardware-Aware Convex Method and Supply Voltage Modulation,” *2025 KIEES Winter Conf.*, Feb. 2025.
- [2] **Y.-S. Lee**, S. Nam, and J. Oh*, “RF-to-DC Efficiency Enhancement in Wireless Power Transfer Systems via Convex Optimization,” *2024 KIEES Summer Conf.*, Aug. 2024.
- [1] H.Y. Kim, **Y.-S. Lee**, and S. Nam*, “Multi-Receiver Charging Algorithm in Microwave Wireless Power Transfer,” *2021 KIEES Autumn Conf.*, Nov. 2021.

AWARDS

Conference Awards - International

- IEEE WPTCE 2026, **Best Student Paper Award (Sole Recipient(1st Place), \$800 USD)**, (First author), Jul. 2026.
- ISAP 2024, **Best Paper Award (2nd Prize)**, (First author), Nov. 2024.

Conference Awards - Domestic

- KIEES Winter Conference 2026, **Best Paper Award (IEEE MTT-S Seoul Chapter Award)**, (Second author), Feb. 2026.
- KIEES Winter Conference 2025, **Best Paper Award (Grand Prize(1st Place), ₩1M KRW)**, (First author), Feb. 2025.

Grants & Scholarships

- IEEE WPTCE 2026, **IEEE MTT-S Student Travel Grant (\$1,000 USD)**, Jul. 2026.
- IEEE WPTCE 2025, **Ph.D. Students Initiative Program (\$450 USD)**, Jun. 2025.
- IEEE WPTCE 2024, **Student Travel Grant (¥100k JPY)**, May 2024.

Student Paper Competitions

- 2024 IEEE Seoul Section Student Paper Contest, **Bronze Prize**, (Second author), Dec. 2024.
- KCA 23rd Student Paper Competition in Propagation, **Excellence Award**, (Second author), Nov. 2024.
- KIEES 9th Electromagnetic Measurement Paper Competition, **Excellence Award**, (First author), Mar. 2024.

Invited Papers

- 2025 Asia-Pacific Microwave Conference (APMC 2025), **Invited Paper Presentation**, Dec. 2025.

SCHOLARSHIPS

- **Ph.D. Scholarship Recipient**, Samsung Electronics Device Solution(DS), Foundry Division (Sep. 2025 -)
- **Teaching & Research Support Scholarship**, Seoul National University (Feb. 2023)
- **Academic Excellence Scholarship**, Seoul National University (Jan. 2022)

REVIEWER ACTIVITIES

IEEE Journals

- IEEE Transactions on Power Electronics (TPEL): 2025, 2026.

PATENTS

[2] (Korea Patent) METHOD FOR DETERMINING OPTIMAL PHASES OF ANTENNA ARRAY IN RECONFIGURABLE INTELLIGENT SURFACE AND RIS SYSTEM USING THE SAME

- Inventors: S. Nam and **Y.-S. Lee**
- Patent No.: 1027047370000
- Granted date: 2024.09.04

[1] (Korea Patent) METHOD FOR DETERMINING OPTIMAL PHASES OF ANTENNA ARRAY IN TRANSMISSION ANTENNA AND A BEAM FOCUSING ARRAY SYSTEM USING THE SAME

- Inventors: S. Nam and **Y.-S. Lee**
- Patent No.: 1026925510000
- Granted date: 2024.08.01

PROJECTS

- [P5] Calibration of Ultra-Large-Scale Digital Array Antennas** Apr. 2026 - Mar. 2028
LIG D&A *Yongin-si, South Korea*
- **Keywords:** *Space Surveillance Radar, Ultra-Large-Scale Digital Array Antenna, Fixed Probe Calibration*
 - Calibration and verification algorithms for ultra-large-scale digital array antennas (86,400+ elements) used in space surveillance radar
 - AI-integrated calibration verification simulator enabling independent and combined sensitivity analysis of error sources
- [P4] Large-Area Wireless Power Transfer System for Power Gateway Application** Mar. 2023 - Mar. 2024
Samsung A-Lab., Samsung Advanced Institute of Technology (SAIT) *Suwon-si, South Korea*
- **Keywords:** *RF-Wireless Power Transfer, Array antenna, RF System, High power*
 - Large scale WPT system design and development (@5.64 GHz, Tx 32x32, Rx 16x16)
 - High power system algorithm development (transmit RF power 1000W+)
- [P3] High-Efficiency BCE 2D Array RF System Based on Time Reversal** Mar. 2023 - Nov. 2023
Samsung A-Lab., Samsung Advanced Institute of Technology (SAIT) *Suwon-si, South Korea*
- **Keywords:** *Time-Reversal, RF-Wireless Power Transfer, Array antenna, RF System*
 - WPT system design and fabrication (@5.64 GHz, Tx 16x16, Rx 7x5)
 - System operation software design and development
 - Large array antenna calibration method design and implementation
- [P2] MIMO Wireless Power Transfer Using Hybrid Beamforming** Mar. 2023 - Sep. 2023
Samsung Device Solution (Samsung DS) *Hwaseong-si, South Korea*
- **Keywords:** *Feedback based algorithms, RF-Wireless Power Transfer, Array antenna, RF System*
 - Received power feedback based RF-WPT optimization algorithm
- [P1] Wireless Power Transfer System Design** Sep. 2021 - Aug. 2022
Samsung Research (SR) *Seoul, South Korea*
- **Keywords:** *RF-Wireless Power Transfer, Array antenna, mmWave antenna, RF System, Link-budget*
 - mmWave large array transmitter antenna (@24 GHz, 32x32)
 - WPT system link budget with power efficiency

RESEARCH EXPERIENCES

- Seoul National University** Jan. 2021 - Aug. 2021
Undergraduate Intern | Advisor: Prof. Sangwook Nam *Seoul, South Korea*
- Applied Electromagnetics Laboratory (AEL)
 - RADAR system used DC to DC Switch Circuit(50W) PCB design & artwork
 - Trained RF design tools (Cadence OrCAD, ADS, CST Studio)
- Korea University** Mar. 2020 - Jun. 2020
Undergraduate Intern | Advisor: Prof. Chulwoo Kim *Seoul, South Korea*
- Advanced Integrated Systems Laboratory (AISL)
 - Bachelor's Thesis: "High-speed 2-Stage Cascode Op-amp design"
 - Cascode Op-amp, VCO, PLL design trainee
 - Trained circuit design tools (Cadence Virtuoso)
- Korea University** Dec. 2019 - Feb. 2020
Undergraduate Intern | Advisor: Prof. Hyung-min Lee *Seoul, South Korea*
- Bio-inspired Advanced System and Integrated Circuit Laboratory (BASIC Lab.)
 - Conventional 2-Stage amp design trainee
 - Trained circuit design tools (Cadence Virtuoso)

SELECTED COURSEWORKS

- Probability and Random Process
- Electromagnetics
- Electromagnetic Fields
- Signal and Systems
- Data Structure & Algorithms
- Digital Signal Processing
- Analog Communication Theory
- Digital Communication Theory
- Wireless Networks
- Electromagnetic Wave Engineering *
- RF System Design *
- Advanced Electromagnetics I *
- Electromagnetic Wave Engineering *
- Introduction to Artificial Intelligence *
- Machine Learning & Deep Learning *
- Electro-optics *
- Estimation Theory *
- Advanced Electricity Market Theory *

*Symbol * denotes graduate course*

UNDERGRADUATE CLASS TA

- Fall 2025, *Introduction to Electromagnetism with Practice*, Seoul Nat'l Univ. (430.202B 003)
- Fall 2024, *Introduction to Electromagnetism with Practice*, Seoul Nat'l Univ. (430.202B 002)
- Spring 2023, *Electromagnetic Wave Engineering*, Seoul Nat'l Univ. (430.464A 001)